MTH 211 : Theory of Statistics

Lecture 1

Revision of topics 2 (Measures of Central Tendency), 3 (Measures of Dispersion), and 4 (Measures of association) from old FCH.

I. Measures of central Tendency : Mean, Median, Mode

Mean is further classified as Arithmetic, Geometric & Harmonic Means.

For any set of positive real numbers : $AM \geq GM \geq HM$, $AM \times HM = (GM)^2$

For positive real nos, $x_1, \ldots, x_n$, $AM = \frac{1}{n}\sum_1^n x_i$, $GM = \left(\prod_1^n x_i\right)^{1/n}$,

$$H.M = \frac{n}{\sum_1^n 1/x_i}$$

For positively skewed data Mean > Median > Mode (unimodal data)

negatively Mean < Median < Mode

symmetric Mean = Median = Mode

For moderately skewed data, Mean − Median $\simeq \frac{1}{3}$ (Mean − Mode)

The above are with respect to observed data.

II. For a random variable X following a probability model f_θ (a prob. distribution) (characterised by parameter/s $\theta / \underset{\sim}{\theta}$),

the rth order raw moment (population) $\mu_r' = E(X^r)$

If $X_1, \ldots, X_n$ is a random sample from this population,

then the rth order raw moment (sample) $m_r' = \frac{1}{n}\sum_1^n X_i^r$

Of special significance is $\mu_1' = E(X) = \mu$, say (more common notation)

This is the population mean

$m_1' = \frac{1}{n}\sum X_i = \bar{X}$ is the sample mean

The rth order central moment $\mu_r = E(X-\mu)^r$. This is the population rth order central moment. The rth order (sample) central moment $m_r = \frac{1}{n}\sum(X_i - \bar{X})^r$.

Of special significance is $\mu_2 = E(X-\mu)^2 = \sigma^2$, say, variance of the distribution,

Consequently, $m_2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ is the sample variance.

We define <u>moment generating function</u> of the r.v. X as
$M_X(t) = E(e^{tX})$

If the mgf exists and is finite $\forall t$ in some $(-a, a)$, then mgf uniquely determines the cumulative distribution fn (cdf).

We can obtain the raw moments from the mgf, viz.,

$$\mu_r' = E(X^r) = \frac{d^r}{dt^r} M_X(t)\Big|_{t=0}$$ (for both discrete and continuous r.v.s)

Some other features involving moments

<u>Skewness (γ_1)</u>, $\gamma_1^2 = \frac{\mu_3^2}{\mu_2^3}$, $\gamma_1 = \frac{\mu_3}{(\sigma^2)^{3/2}}$

and $$g_1 = \frac{\frac{1}{n}\Sigma(x_i - \bar{x})^3}{\left(\frac{1}{n}\Sigma(x_i-\bar{x})^2\right)^{3/2}}$$

All odd order central moments of a symmetric are equal to 0.

Hence, for a symmetric distribution $\gamma_1 = 0$ (normal, t, Cauchy, Laplace ...)
positively skew $>$ (exponential, Gamma, ...
negatively skew $<$ (spl. values of parameters leading to negatively skew)

<u>Kurtosis</u> $\gamma_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$, Excess kurtosis $\gamma_2 - 3$

For $X \sim N(\mu, \sigma^2)$, $\mu_{2r} = (2r-1)(2r-3)\cdots 3.1\, \sigma^{2r}$ (Check!), so that
$\gamma_2 = 3$ and excess kurtosis $= 0$ (mesokurtic)
$>$ for leptokurtic distributions
$<$ platykurtic distributions.

Measures like Mean Absolute Deviation $E|X-\mu|$, Variance $E(X-\mu)^2$, etc. have special significance as loss function.

The coefficient of variation (c.v.) $= \frac{\sigma}{\mu}$ measures dispersion wrt mean & has significance in interval estimation.

<u>Population quantiles</u>, $\xi_p \ni P(X \le \xi_p) = p$, $0 < p < 1$;
ξ_p is the <u>pth</u> population quantile

Of special significance, as $p = \frac{p^*}{100}$, $1 \le p^* \le 99$ (percentiles)
$= \frac{p^*}{10}$, $1 \le p^* \le 9$ (deciles)
$= \frac{p^*}{4}$, $1 \le p^* \le 3$ (quartiles)

If $X_1, \dots, X_n$ is a random sample and $X_{(1)} \le \dots \le X_{(n)}$ the corresponding order statistics

Then $X_{(r)}$ with $r = np$ if $np \in I$, $[np]+1$ if $\notin$, is the corresponding <u>sample quantile</u>.

III <u>Measures of Association</u> (bivariate)

$X, Y \sim F_{X,Y}$

n bivariate sample obsns $(x_1, y_1), \dots, (x_n, y_n)$ from $F_{X,Y}$.

ⓐ <u>Pearson's correlation coefficient</u> $\rho = \dfrac{\mathrm{Cov}(X,Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$, with the

<u>sample correlation coefficient</u> $r = \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2}\sqrt{\sum(y_i-\bar{y})^2}}$

If X, Y are indep. i.e. $F_{X,Y}(x,y) = F_X(x)F_Y(y)\ \forall (x,y)$ and $\rho = 0$

Converse is not true, but true for $N_2(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho_{XY})$. Why??

ⓑ Let $R_i = \mathrm{rank}(X_i)$, $S_i = \mathrm{rank}(Y_i)$ Obsns are ranked componentwise, $X_{(1)} < \dots < X_{(n)}$ and $Y_{(1)} < \dots < Y_{(n)}$

(Note, $i = \mathrm{rank}(X_{(i)}) = \mathrm{rank}\ Y_{(i)}$).

Then, <u>Spearman's rank correlation</u> $r_s = \dfrac{\sum_i (R_i-\bar{R})(S_i-\bar{S})}{\sqrt{\sum(R_i-\bar{R})^2}\sqrt{\sum(S_i-\bar{S})^2}}$

$= 1 - \dfrac{6\sum d_i^2}{n(n^2-1)}$, $d_i = R_i - S_i$ (Check!)

<u>Kendall's</u> $\tau = \pi_c - \pi_d$, where

between any 2 pairs (x_i, y_i) and (x_j, y_j)

Probability of concordance $\pi_c = P[(X_i - X_j)(Y_i - Y_j) > 0]$

$= \frac{1}{2}$ under independence and absolutely continuous marginals F_X and F_Y (why?)

Probability of discordance $\pi_d = P[(X_i - X_j)(Y_i - Y_j) < 0]$

$= \frac{1}{2}$ (argument as above)

So, $\tau = 0$ if $X \sim F_X$ and $Y \sim F_Y$ (both absolutely conts) are independent

The sample Kendall's coeff. is $T = \dfrac{C - D}{\binom{n}{2}}$ where $C(D) = \#$ concordant (discordant) pairs.

e.g. if $(2,3), (3,5), (4,2)$ are 3 obsns $T = \dfrac{1-2}{3} = -\dfrac{1}{3}$.